

Competitive Exam Practice 6: Hardwired Control Unit Design

GATE (Computer Science) Pattern

PCC CS-401: Computer Organization and Architecture

August 7, 2026

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

- Q1** [*Original, GATE-pattern*] [**MCQ, 1 mark**] In a hardwired control unit, every control signal is generated as a function of:
- (A) Opcode decoder lines only
 - (B) Step decoder lines only
 - (C) Both the opcode lines and the step lines together
 - (D) Neither — signals are read from a stored table
- Q2** [*Original, GATE-pattern*] [**NAT, 1 mark**] An instruction set architecture has 25 distinct instructions. The minimum number of bits required in the instruction's opcode field to encode all of them is _____.
- Q3** [*Original, GATE-pattern*] [**MCQ, 1 mark**] A sequence counter (SC) together with a step decoder, as used to generate one-hot control-step signals in a hardwired control unit, is functionally most similar to:
- (A) A program counter driving an instruction cache
 - (B) An instruction register holding the current opcode
 - (C) A modulo- n counter feeding an n -output decoder
 - (D) A stack pointer
- Q4** [*Original, GATE-pattern*] [**NAT, 1 mark**] A hardwired control unit's sequence counter is implemented with 6 bits. The maximum number of distinct control steps it can address is _____.
- Q5** [*Original, GATE-pattern*] [**MCQ, 2 marks**] Which of the following is **NOT** a disadvantage of hardwired control relative to microprogrammed control?
- (A) Difficult to modify after the chip is fabricated
 - (B) Design/verification complexity grows with instruction-set size
 - (C) Slower per control step than microprogrammed control
 - (D) Requires re-deriving Boolean equations whenever the ISA changes
- Q6** [*Original, GATE-pattern*] [**MCQ, 2 marks**] For a control signal shared by two instructions at the same control step, e.g. $Z_{in} = (D_{ADD} \cdot T_2) + (D_{SUB} \cdot T_2)$, the OR operation reflects the fact that:
- (A) ADD and SUB must execute simultaneously

- (B) Z_{in} must be asserted whenever *any* instruction whose state table lists it at that step is running
- (C) Z is a shared bus line, not a register
- (D) The equation is redundant and can be simplified to a single term